Crown Corking Machine



Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Crown corking machine

What it is used for

A crown corking machine seals compatible glass bottles using metal crown caps to create a secure closure. Correct cap and bottle matching, controlled capping pressure, hygienic handling, and closure inspection help reduce leakage, contamination, and breakage.

Main parts

- Machine frame supports the capping assembly.
- Capping head presses the crown cap onto the bottle finish.
- Cap holder or magnetic head positions the crown cap where fitted.
- Bottle support or platform holds the bottle upright.
- Height adjustment accommodates approved bottle sizes.
- Lever, pneumatic actuator, or motor provides capping force.
- Control switch or foot pedal operates powered models.
- Safety guard protects against moving parts where fitted.
- Crown caps provide the bottle closure.

Before you start

1. Only trained personnel should operate the crown corking machine.
2. Read the manual and confirm that the crown caps and bottle finish are compatible with the machine.
3. Inspect the frame, capping head, bottle platform, controls, guards, and drive components for damage.
4. Clean and sanitise bottle-contact surfaces according to the approved hygiene procedure.
5. Check bottles for cracks, chips, contamination, and damaged neck finishes; reject defective bottles.
6. Confirm that caps are clean, undamaged, and suitable for the intended beverage and closure requirements.
7. Adjust the capping height and pressure according to the machine instructions and bottle specifications.

How to use

1. Place the machine on a stable surface and prepare the required bottle and crown cap.
2. Position the bottle upright on the designated support and align its neck under the capping head.
3. Place the crown cap squarely on the bottle mouth using the approved cap-loading method.
4. Keep hands clear of the capping head and activate the lever, pedal, or control as designed.

5. Apply the capping cycle until the cap is fully seated and the machine returns to its safe position.
6. Remove the bottle carefully and inspect the cap for uniform seating, crimping, and visible damage.
7. Check for leakage or closure defects using the approved quality-control method.
8. Repeat the process while monitoring cap placement, machine performance, and bottle condition.
9. Stop the machine if a bottle cracks, a cap jams, or the capping mechanism behaves abnormally.
10. At the end of the run, switch off and isolate power before clearing jams or cleaning the machine.
11. Clean and sanitise appropriate surfaces and record inspection results according to production procedures.

⚠ Safety

- Keep fingers away from the capping head, press mechanism, and bottle support during operation.
- Do not use cracked or chipped bottles because they may break under capping force.
- Never bypass guards or reach into a powered machine to clear a jam.
- Use the specified cap type and machine settings; excessive force may damage bottles or closures.
- Handle broken glass with suitable tools and protective equipment, never with bare hands.
- Follow beverage hygiene requirements and prevent contamination of bottle mouths and caps.

Care & storage

- Clean and sanitise bottle-contact surfaces after production using approved methods.
- Inspect the capping head, cap holder, adjustment mechanism, guards, and fasteners regularly.
- Remove cap fragments and broken glass only after isolating the machine and following site procedures.
- Keep the machine dry and protect it from corrosion and impact.
- Arrange authorised servicing if cap seating becomes inconsistent or the mechanism wears.

Common mistakes to avoid

- Using caps that do not match the bottle finish or machine head.
- Misaligning the bottle or cap before applying pressure.
- Applying excessive force and cracking the bottle neck.
- Failing to inspect closures or safely remove broken glass.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.